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Explainable AI for Energy Prediction and Anomaly Detection in Solar Energy Systems

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Abstract

The integration of solar energy systems into power grids is gaining momentum globally, promising sustainable and renewable energy solutions. However, the intermittent and dynamic nature of solar energy poses challenges for its efficient utilization and management. In response, this research investigates the application of Explainable Artificial Intelligence (XAI) techniques for enhancing the prediction accuracy and anomaly detection in solar energy systems. By leveraging advanced machine learning algorithms and interpretable models, this study aims to provide insights into the underlying factors influencing solar energy generation, thus enabling better decision-making and optimization strategies. In order to make machine learning models more understandable, approaches such as Explainable Artificial Intelligence (XAI) have been developed. The understanding and confidence that are ingrained in the output of machine learning algorithms is what makes XAI significant. In order to forecast Key Performance Indicators (KPIs), the current work constructs a thorough XAI model and evaluates the effectiveness of several learning algorithms using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2) coefficient of determination, the number of iterations, and execution time. The proposed framework holds significant potential to improve the reliability, efficiency, and performance of solar energy systems, contributing to the transition towards a more sustainable and resilient energy infrastructure.

1 Introduction

Over the past century, energy has drastically changed both human lives and the geopolitics of planet [1], [2]. Examining the past and present world conflicts, energy has played a pivotal role in numerous instances, encompassing resources such as oil, natural gas, and minerals used in making batteries. The availability of energy paved the way for contemporary technical developments, such as the widespread use of computers and power sources like batteries, and made human being civilizations more intelligent. Conventional electrical power grids have historically experienced operational instability, inefficiency, rigidity, and unreliability [3]. Through its many forms—heat derived from sunlight, photovoltaic energy powered by the sun, warm power derived from sunlight, and fuels oriented toward the sun—solar energy provides humanity with an ideal, abundant, and limitless energy resource. Solar power is the process of converting solar light into electrical current, either directly through photovoltaic (PV) panels or indirectly through concentrated solar power (CSP) panels [4]. The global shift towards renewable energy sources, particularly solar power, underscores the urgency of developing innovative approaches to optimize energy generation, distribution, and consumption. With its abundance and environmental friendliness, solar energy presents a viable way to slow down global warming and lessen reliance on fossil fuels. [5], [6]. However, the inherently variable nature of solar irradiance, influenced by factors such as weather conditions, time of day, and geographic location, presents challenges for harnessing its full potential. As a result, the effective management and operation of solar energy systems require sophisticated predictive modeling and anomaly detection techniques to anticipate fluctuations in energy output and identify anomalous behaviors.

Artificial intelligence (AI) is beginning to play a bigger role in profoundly influencing our daily lives. Furthermore, the effects of AI on individuals and professionals are widespread as AI-based arrangements proliferate in industries including lending, law enforcement, healthcare, and education. AI models play such a large role in several domains that there is growing concern about potential biases in them as well as the need for model interpretability. In high-stakes fields requiring unwavering quality and security, like medical services and automated transportation, as well as in fundamental contemporary applications with substantial financial ramifications, like predictive maintenance, routine asset investigation, and environmental change display, model logic is also essential for fostering trust and acceptance of AI frameworks [7]. Deep learning is responsible for a large amount of the recent advances in artificial intelligence. Deep learning algorithms performed noticeably better than conventional machine learning techniques. Deep Neural Networks (DNNs), on the other hand, are viewed as a black box by both developers and users since they are poor at explaining their inference procedures and outputs [8], [9].

Transparent and comprehensible insights into model predictions are provided by Explainable Artificial Intelligence (XAI), which has become an essential instrument in bridging the gap between sophisticated machine learning algorithms and human decision-makers. In the context of solar energy systems, XAI holds the potential to enhance prediction accuracy, facilitate real-time monitoring, and enable proactive anomaly detection. By providing understandable explanations for model outputs, XAI empowers stakeholders to trust and utilize AI-driven insights effectively, thereby fostering informed decision-making and operational efficiency.

Azeem et.al., have proposed a framework for efficiently managing renewable hybrid AC-DC microgrids, tackling the complexities of modern power systems. The literature covers the topics such as the transformation of traditional distribution systems into active grids, optimal energy management techniques, integration of electric vehicles, cybersecurity measures, the Bat Optimization Algorithm, and Digital Twin Technology [10]. Amit Dhoke et. Al., have addressed the pressing need for automated approaches to safeguard PV systems against faults. It begins by noting the remarkable growth of solar PV installations worldwide and the necessity for robust fault detection and monitoring mechanisms, especially in large-scale PV plants. Challenges in fault detection arise due to the vast area covered by solar farms and the limitations of existing protection devices [11]. Previous studies have explored various fault detection techniques, but there remains a gap in automated fault diagnosis, particularly at the module level. Obando et. al., have addressed the importance of solar radiation prediction and the potential benefits it offers across various domains such as solar energy production, climate research, and agricultural planning. While machine learning algorithms have been utilized for this purpose, there's a gap in exploring explainable AI (XAI) methods [12]. This research focuses on investigating ensemble methods within the framework of XAI to enhance solar radiation prediction. Sarp et.al., have described the widespread adoption of artificial intelligence (AI) across industries, the opaque nature of AI systems remains a challenge, hindering their broader acceptance. However, the interpretability of these forecasts is crucial for enhancing efficiency and fostering further adoption of PV energy. The study presents a use case of PV energy forecasting employing an XAI tool on a high-resolution dataset [13].

The purpose of this research is to investigate how Explainable AI approaches can be used in solar energy systems for anomaly detection and energy prediction. By integrating XAI methods with advanced machine learning algorithms, such as neural networks, decision trees, and ensemble methods, this study seeks to develop a comprehensive framework for optimizing solar energy generation and improving system reliability. Through the analysis of historical energy data, weather patterns, and environmental factors, the proposed approach aims to identify patterns, trends, and anomalies that influence solar energy output, thereby enabling proactive maintenance, fault detection, and performance optimization.

2 Methodology

Data Collection:

The data utilized in this model has been collected directly from an ESP32 microcontroller. This microcontroller was responsible for gathering sensor data and solar panel readings, transmitting them wirelessly to the internet via Wi-Fi. The collected information was then stored and managed on ThingSpeak, an IoT platform. Subsequently, the data was extracted for further analysis and utilized in the development of the solar energy prediction and anomaly detection model.

Data Preprocessing:

To get rid of outliers, inconsistent data, and missing values, clean up the raw data. Scale or normalise the data to make sure everything is consistent and to help with model training. To extract pertinent features, like the time of day, day of the week, sea-sonality, and weather patterns, use feature engineering.

Feature Selection:

Utilize statistical techniques, domain knowledge, and correlation analysis are to identify significant features for predicting solar energy generation.

Model Development:

Use a range of machine learning techniques, such as recurrent neural networks (RNNs) or long short-term memory (LSTM) networks, to capture temporal dependencies and sequential patterns. These techniques are appropriate for time-series prediction and anomaly detection. For robust prediction and ensemble learning, use Gradient Boosting Machines (GBMs), Random Forests, or Decision Trees. Utilising the pre-processed data, train the models and optimise the hyperparameters using grid search and cross-validation techniques. To improve the interpretability and transparency of model predictions, incorporate Explainable AI techniques like feature importance analysis, LIME (Local Interpretable Model-agnostic Explanations), and SHAP (SHapley Additive exPlanations) values. The explainable AI flow diagram for a solar cell is displayed in Figure 1.

Model Evaluation:

To evaluate the performance of the model, divide the dataset into test, validation, and training sets. For regression tasks, use appropriate evaluation metrics such as R-squared (R^2) coefficient of determination, Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE).

Mean Absolute Error (MAE): One statistic used to evaluate regression models is the Mean Absolute Error (MAE). The prediction error is the discrepancy for each instance between the true value and the expected value.

$$mae = (\sum_{i=1}^{n} abs(yi - \boldsymbol{\lambda}(\mathrm{xi}))) \div \mathrm{n}$$

Root Mean Squared Error (RMSE): In AI, having a single number to measure a model's performance is very helpful, whether it's for planning, cross-validation, or post-organization verification. The root mean square error is one of the most common estimation errors for this. It's an easy scoring decision that works with some of the most common quantifiable assumptions as well.

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N}(y - \hat{y}_1)^2}{N}}$$

R-squared (R^2): A factual metric called R-squared (R2) assesses the degree of variation explained by an independent variable or by components in a multivariate model for a dependent variable.

$$R^2 = 1 - \frac{\sum(y - \hat{y}_1)^2}{\sum(y - \tilde{y}_1)^2}$$

Interpretation and Visualization:

Interpret the model predictions and XAI explanations to gain insights into the underlying factors influencing solar energy generation. Visualize the results using plots, graphs, heatmaps, and interactive dashboards to facilitate understanding and decision-making by stakeholders. Provide intuitive explanations for model predictions and anomalies, highlighting the significance of key features and their impact on energy output.

3 Model configuration

Hardware Configuration:

- Microcontroller: ESP32 Wi-Fi Module Sensors:
- DHT11

- Voltage Sensor
- LDR
- Solar Panel

Data Acquisition System:

Figure 2 is showing the hardware connection between the different components used. Sensor data and solar panel readings were gathered using an ESP32 microcontroller module, which connected them wirelessly to the internet via Wi-Fi. This microcontroller transmitted the data to ThingSpeak, an IoT platform, where it was stored for analysis. After logging, the data was exported for further examination and utilized in the research for solar energy prediction and anomaly detection.

Software Configuration:

Programming Language: Python 3.10.12

XAI Library: SHAP (SHapley Additive exPlanations) Data Analysis Tools:

● Pandas

● NumPy

● Scikit-learn

● Seaborn

● Matplotlib

Development Environment: Google Colab

4 Results and discussion

Model Explain ability

4.1 Linear Regression:

Figure 3, 4 are showing the impact of various factors, number of anomalies detected of the linear regression model. The linear regression model is mainly influenced by the factors Total Minutes passed and Irradiation. The negative SHapley Additive exPlanations (SHAP) value for the Total Minutes Passed implies that as time progressed the model expects a decrease in the voltage output. For higher values of Irradiation, a positive SHAP value can be observed, hence when the sunlight intensity is more, the voltage produced is also more. By analyzing these coefficients, the linear regression model can identify anomalies in solar energy production, such as unexpected deviations in voltage output given the time of

day or observed irradiation levels. Figure 5 is showing the different factors such as total minutes pass, irradiation, humidity and temperature effects.

4.2 Random Forests:

In contrast to linear regression, where "Total minutes passed" predominantly influences predictions, random forest models prioritize "Irradiation" as the primary factor shaping voltage output estimations is showing in Fig. 6,7. While linear regression relies heavily on the progression of time throughout the day to forecast energy production, random forest models weigh irradiation levels more significantly. Consequently, anomalies detected by random forest algorithms stem from deviations between observed voltage and predicted values driven by unexpected variations in sunlight exposure rather than daily time progression. Figure 8 is about the dependency profile with various parameters such as irradiation, humidity for random forest model.

4.3 Gradient Boosting:

Similarly, gradient boosting models exhibit a preference for "Irradiation" over time progression in shaping voltage output estimations. Like random forest models, gradient boosting algorithms assign higher importance to irradiation levels, resulting in similar SHAP values which is showing in Fig. 9, 10. This emphasis on irradiation in both random forest and gradient boosting models underscores the critical role of sunlight exposure in predicting voltage output, with anomalies primarily arising from deviations in irradiation levels rather than the passage of time. Figure 11 is showing the dependency graph for the gradient boosting model.

Model Evaluation:

Models	RMSE	MAE	R2
Linear Regression	0.033	0.022	0.97
Random Forest	0.041	0.024	0.99
Gradient Boosting	0.05	0.03	0.99

The results of our analysis reveal promising outcomes in fault detection and prediction within solar energy production systems. By employing advanced data preprocessing techniques and machine learning algorithms, we achieved accurate predictions of solar power generation, thereby enabling proactive fault detection and mitigation strategies. The trained models exhibited high performance, as indicated by significant R-squared scores, underscoring their effectiveness in optimizing energy production which is showing in Fig. 12. Additionally, our experimental analysis showcases corrected predictions with minimal errors, further validating the reliability of the proposed methodologies.

Conclusion

In conclusion, this research highlights the significance of explainable AI techniques in improving the reliability and interpretability of energy prediction and anomaly detection in solar energy systems. By integrating SHAP values into deep learning models, we enhanced the transparency of predictions, enabling stakeholders to understand the underlying factors influencing energy generation forecasts. Moreover, our anomaly detection framework based on isolation forest offers a robust solution for identifying irregularities in solar energy data, facilitating early intervention and system resilience. The combination of accurate energy prediction and proactive anomaly detection contributes to the overall efficiency and performance of solar energy systems, ultimately leading to cost savings and environmental benefits. Looking ahead, future research could explore the application of additional explainable AI methods, such as LIME (Local Interpretable Model-agnostic Explanations) or integrated gradients, to further enhance the interpretability of predictive models in renewable energy systems. Additionally, investigating ensemble techniques and hybrid approaches for anomaly detection may offer opportunities for even greater accuracy and reliability in identifying system faults and optimizing energy production. Overall, our study underscores the importance of explainable AI in advancing the state-of-the-art in solar energy forecasting and anomaly detection, paving the way for more transparent, reliable, and sustainable energy systems.

Declarations

Author Contribution

All authors wrote and reviewed the manuscript.

Data Availability

Data is provided within the supplementary information files

References

1. M. Blondeel, M. J. Bradshaw, G. Bridge, and C. Kuzemko, “The geopolitics of energy system transformation : A review,” no. June, pp. 1–22, 2021, doi: 10.1111/gec3.12580.
2. R. Vakulchuk, I. Overland, and D. Scholten, “Renewable energy and geopolitics : A review,” *Renewable and Sustainable Energy Reviews*, vol. 122, no. January, p. 109547, 2020, doi: 10.1016/j.rser.2019.109547.
3. R. Alsaigh, R. Mehmood, and I. Katib, “AI explainability and governance in smart energy systems : A review,” no. January, pp. 1–12, 2023, doi: 10.3389/fenrg.2023.1071291.
4. G. K. Singh, “Solar power generation by PV (photovoltaic) technology : A review,” *Energy*, vol. 53, pp. 1–13, 2013, doi: 10.1016/j.energy.2013.02.057.
5. S. Kalogirou, “The potential of solar industrial process heat applications,” vol. 76, pp. 337–361, 2003, doi: 10.1016/S0306-2619(02)00176-9.

6. P. Torino, “KiteGen : A revolution in wind energy generation,” no. June 2020, pp. 2008–2009, 2009, doi: 10.1016/j.energy.2008.10.003.
7. W. Samek, “arXiv : 1909 . 12072v1 [cs . AI] 26 Sep 2019,” vol. 11700, pp. 5–22, 2019.
8. X. A. I. Toward, M. Xai, and E. Tjoa, “A Survey on Explainable Artificial Intelligence,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 32, no. 11, pp. 4793–4813, 2021, doi: 10.1109/TNNLS.2020.3027314.
9. G. Sharma, “Explainable AI for Industry 5 . 0 : Vision , Architecture , and Potential Directions,” *IEEE Open Journal of Industry Applications*, vol. 5, no. December 2023, pp. 177–208, 2024, doi: 10.1109/OJIA.2024.3399057.
10. O. Azeem *et al.*, “applied sciences A Comprehensive Review on Integration Challenges , Optimization Techniques and Control Strategies of Hybrid AC / DC Microgrid,” 2021.
11. A. Dhoke, R. Sharma, and T. K. Saha, “An approach for fault detection and location in solar PV systems,” *Solar Energy*, vol. 194, no. November 2018, pp. 197–208, 2019, doi: 10.1016/j.solener.2019.10.052.
12. E. Obando, S. Carvajal, and J. Pineda, “Learning Techniques : A Review,” vol. 17, no. 4, pp. 684–697, 2019.
13. S. Sarp, M. Kuzlu, and U. Cali, “A Highly Transparent and Explainable Artificial Intelligence Tool for Chronic A Highly Transparent and Explainable Artificial Intelligence Tool for Chronic Wound Classification : XAI-CWC,” no. February, 2021, doi: 10.20944/preprints202101.0346.v1.

Figures

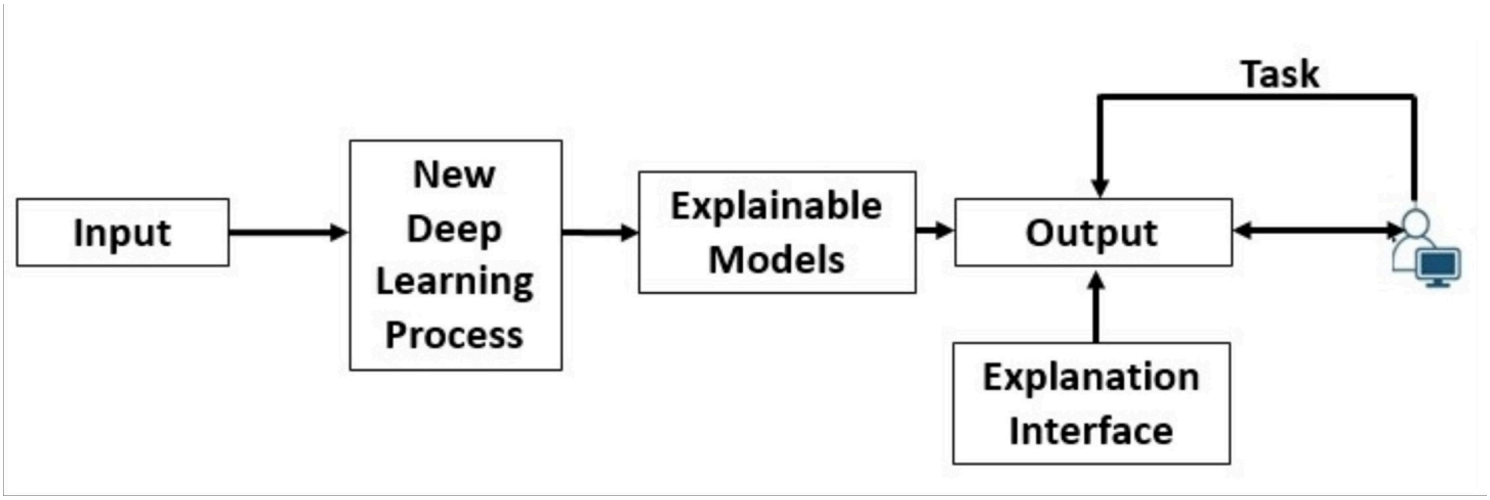


Figure 1

Flow diagram for expandable AI for energy predictions of solar cell.

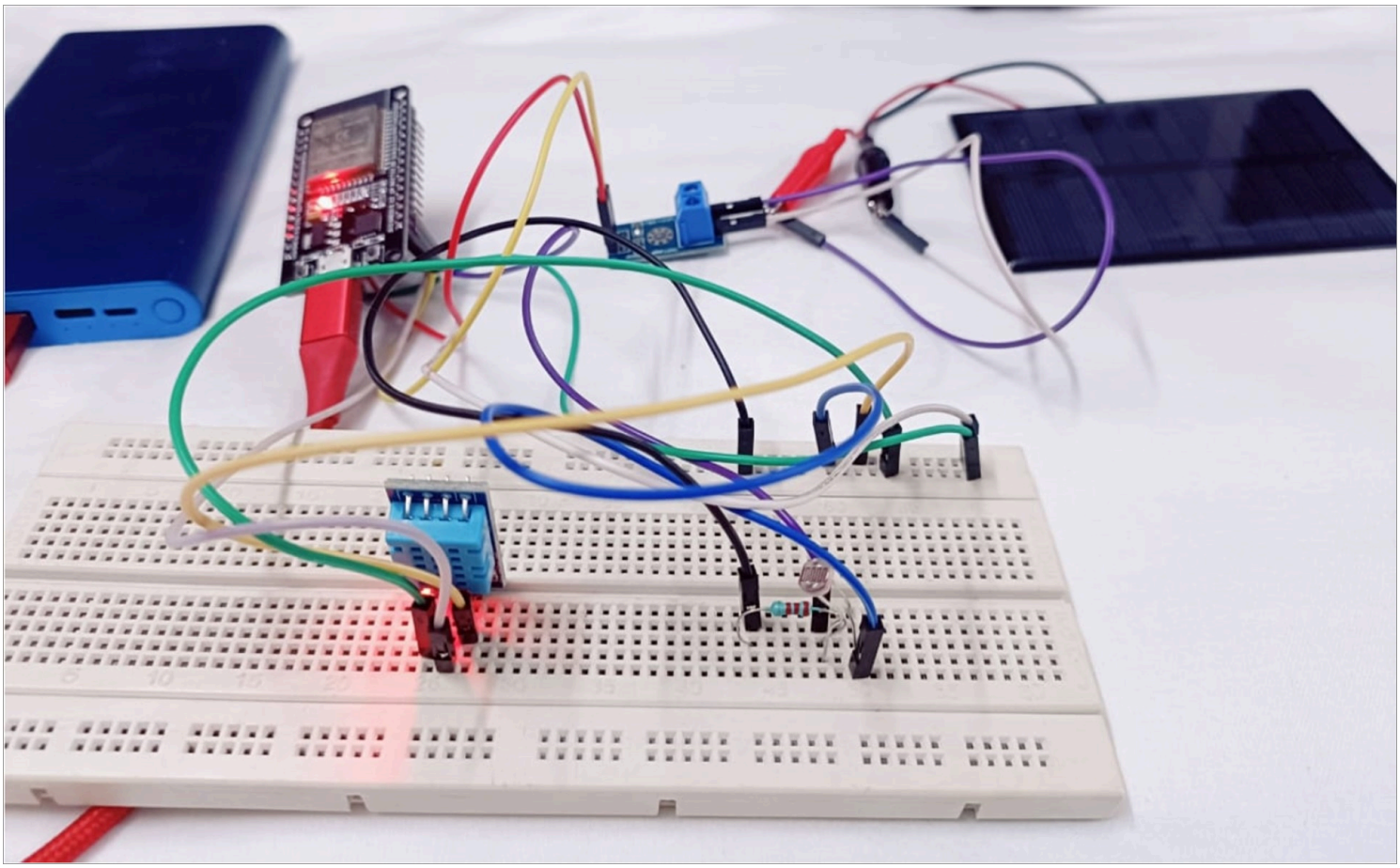


Figure 2

Illustration of Circuit Configuration

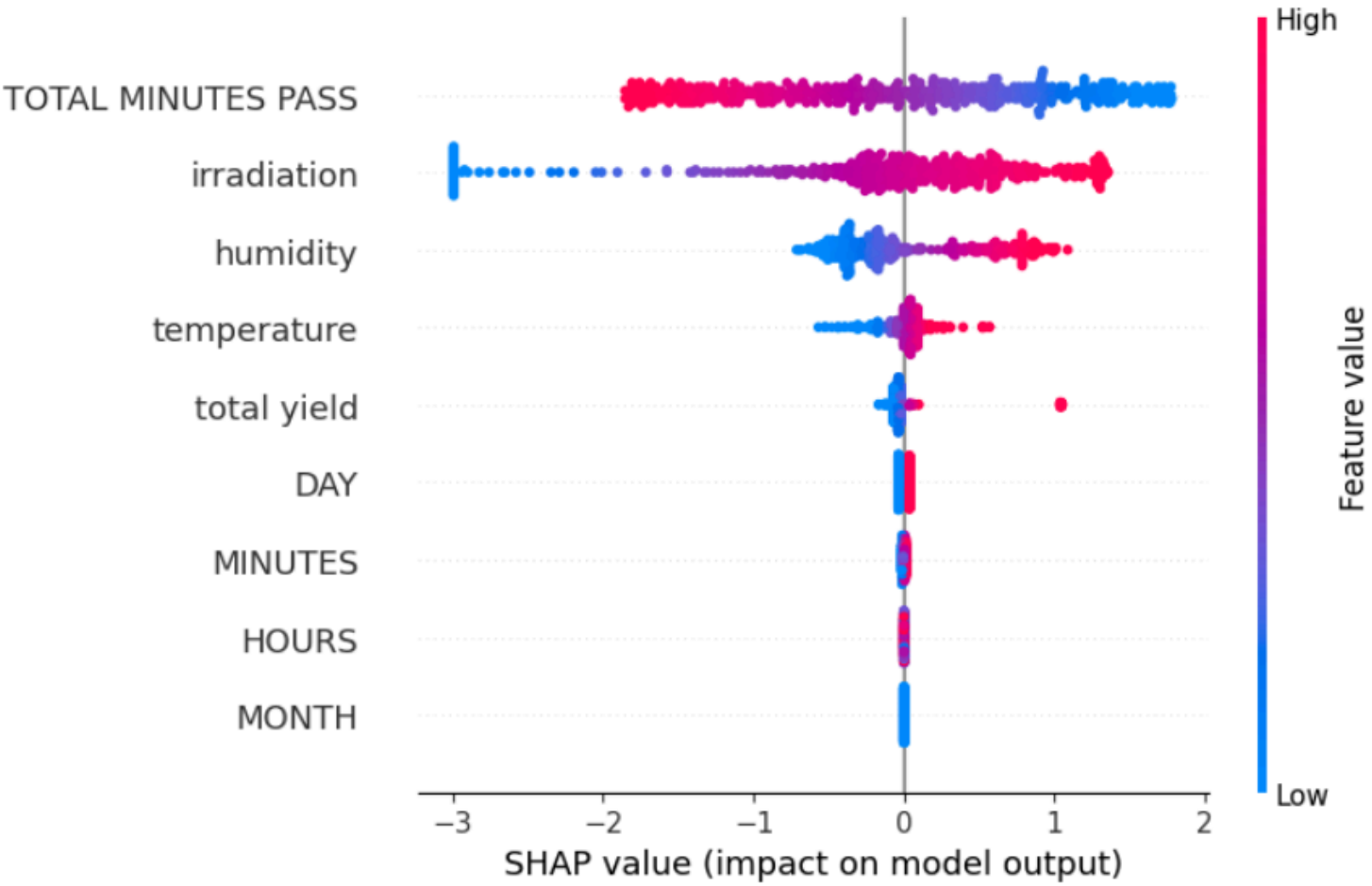


Figure 3

Impact of various factors on linear Regression

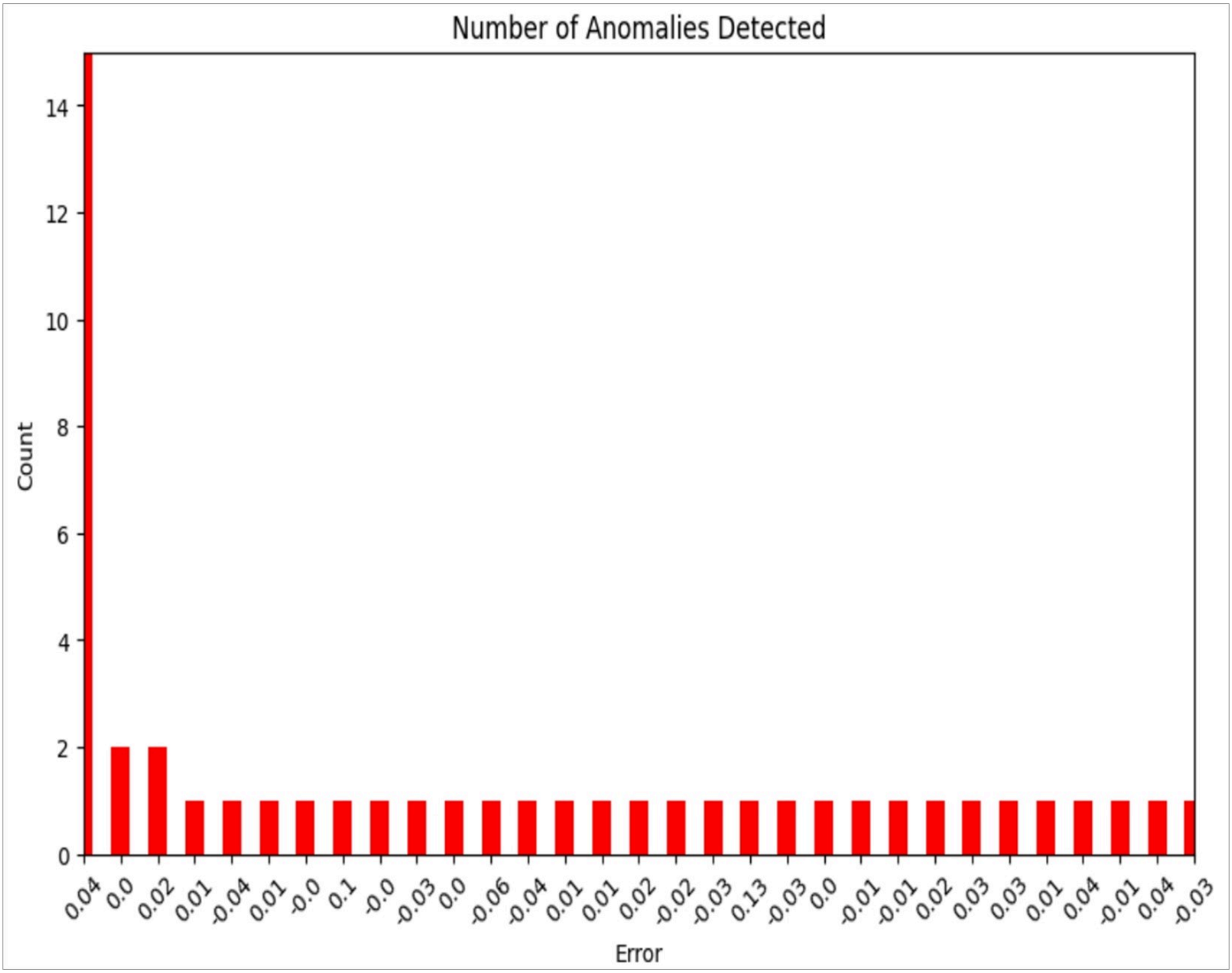


Figure 4

Number of anomalies detected linear Regression

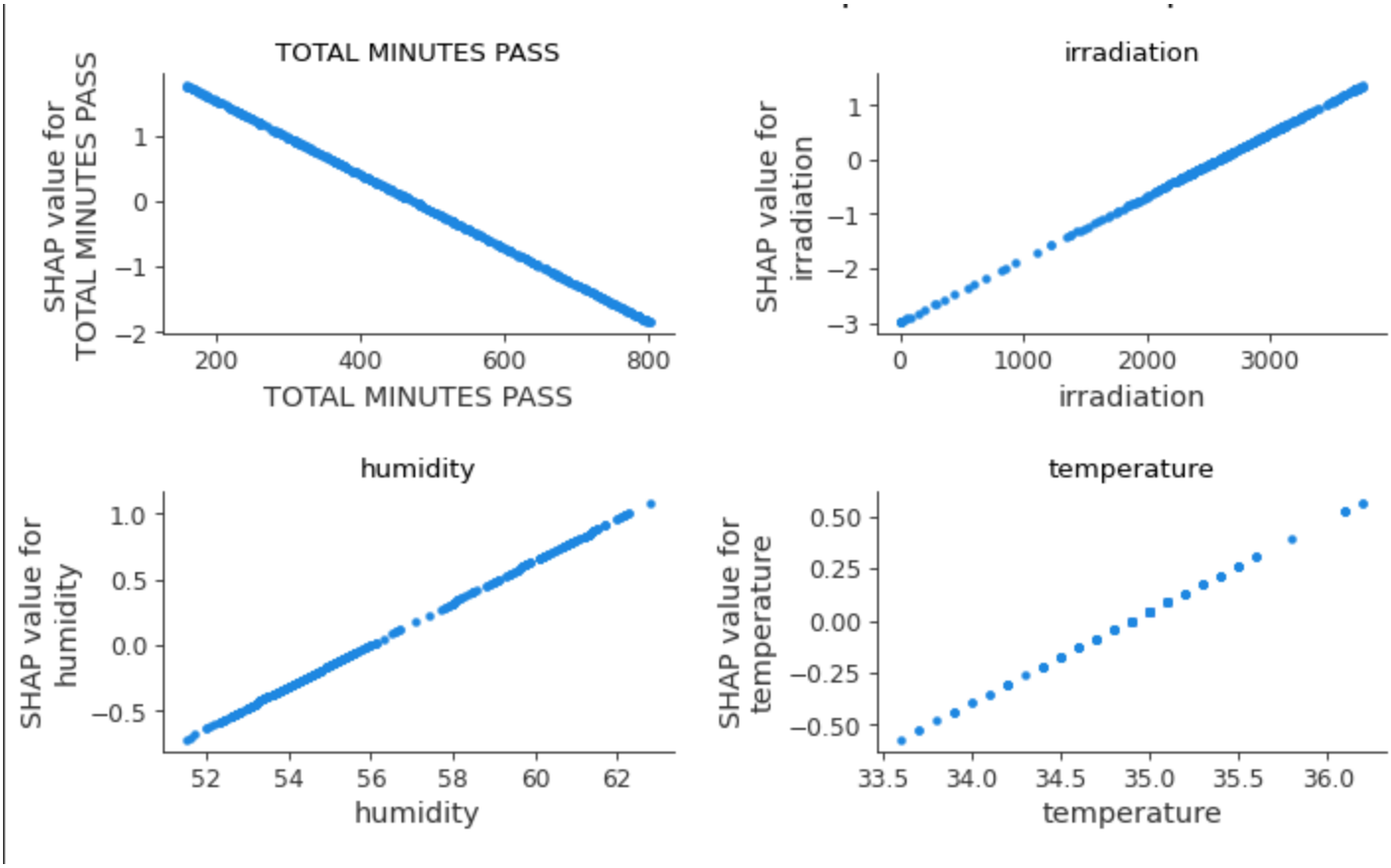


Figure 5

Dependence plot of top 4 factors affecting the prediction on linear regression

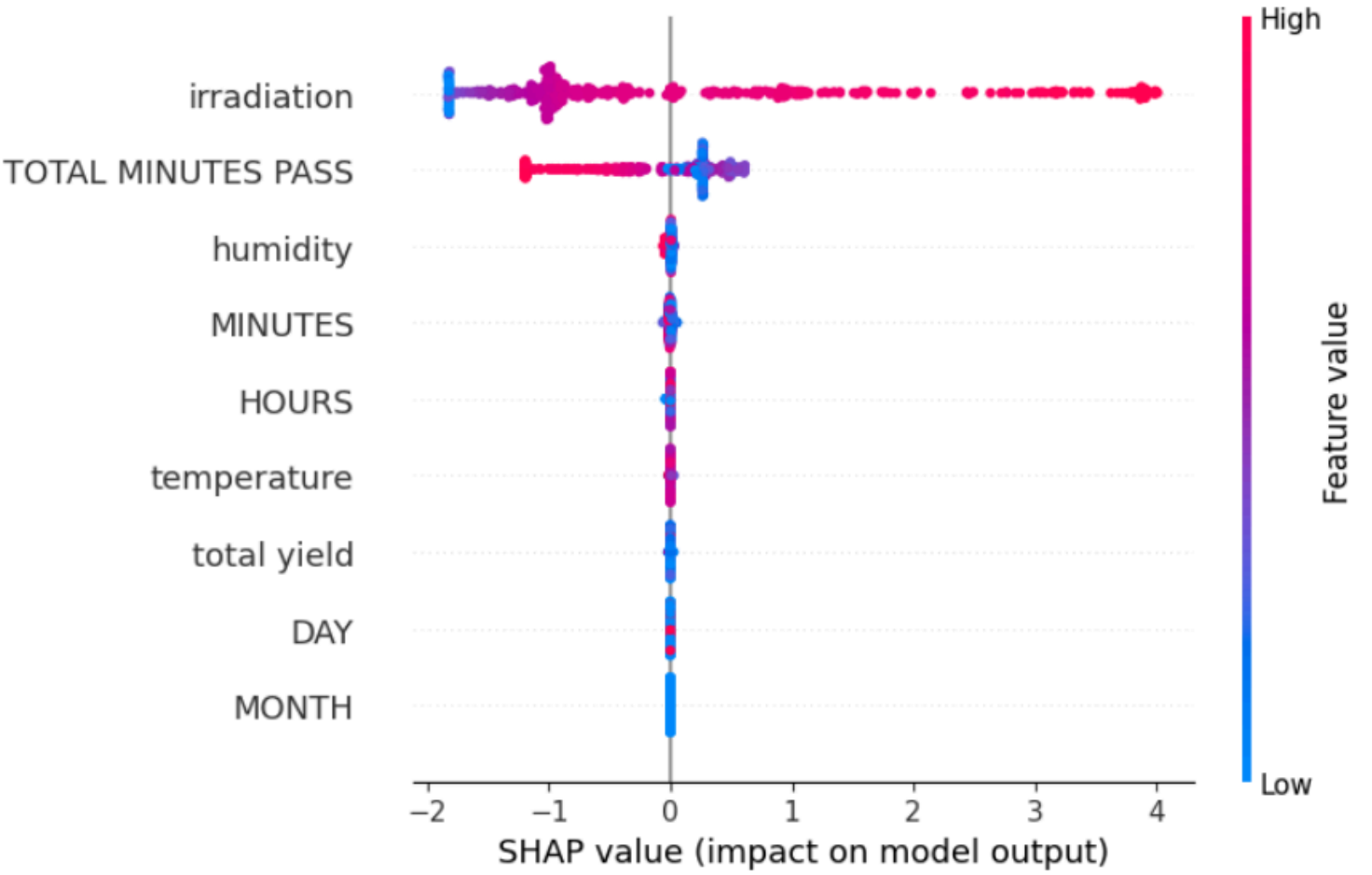


Figure 6

Impact of various factors on the random forest

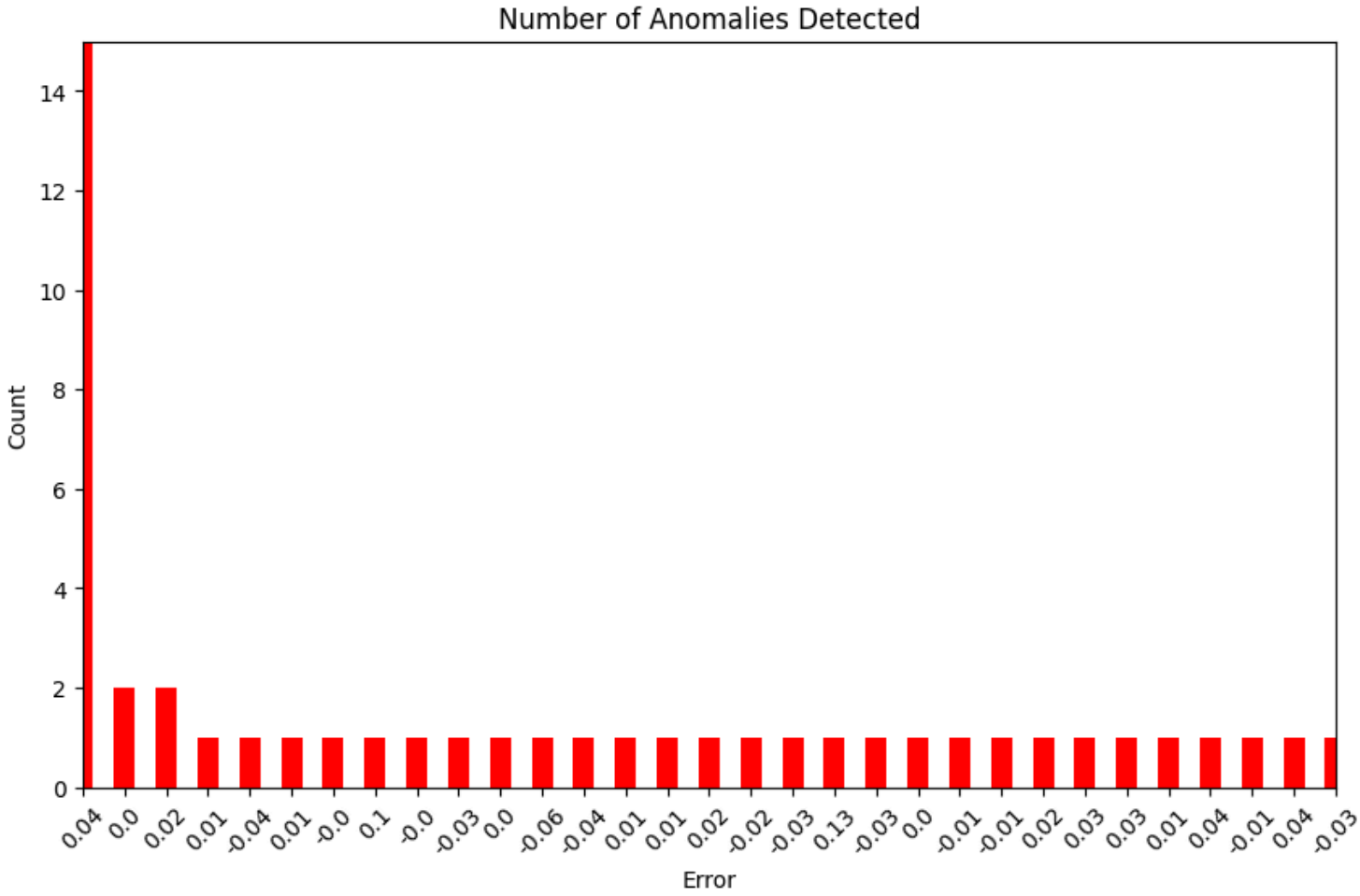


Figure 7

Number of anomalies detected on random forest

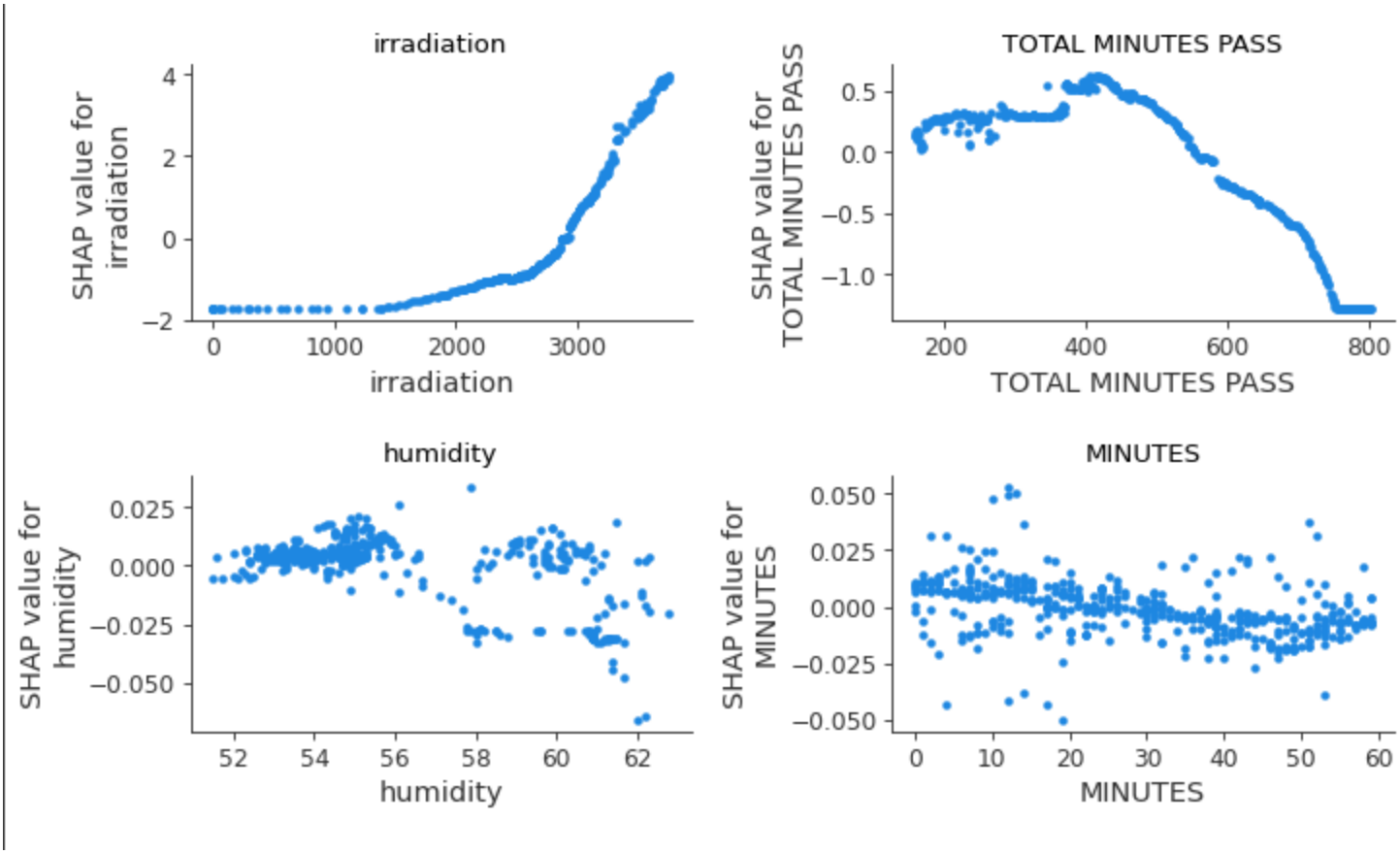


Figure 8

Dependence plot of top 4 factors affecting the prediction on random forest

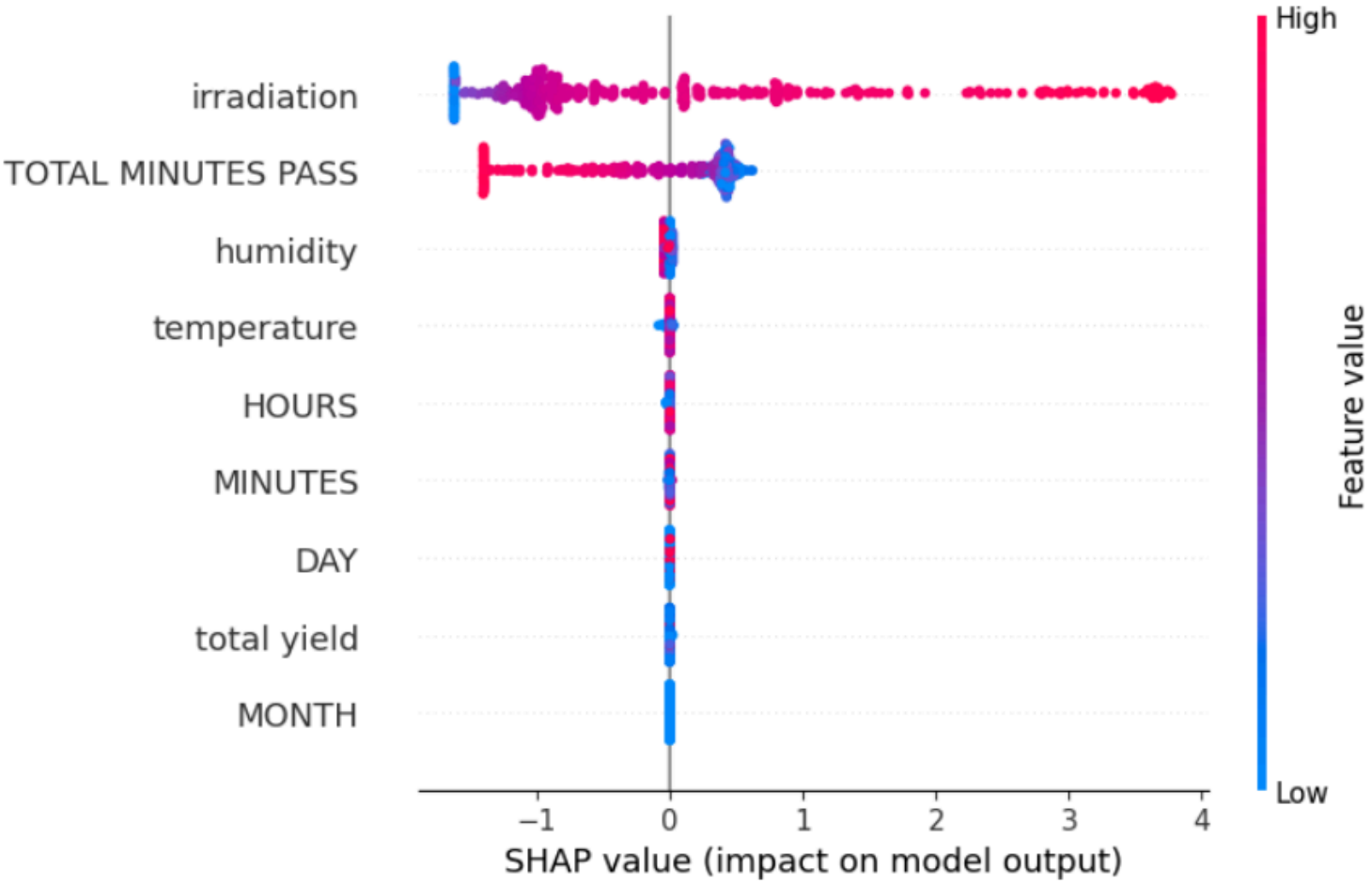


Figure 9

Impact of various factors on the gradient boosting

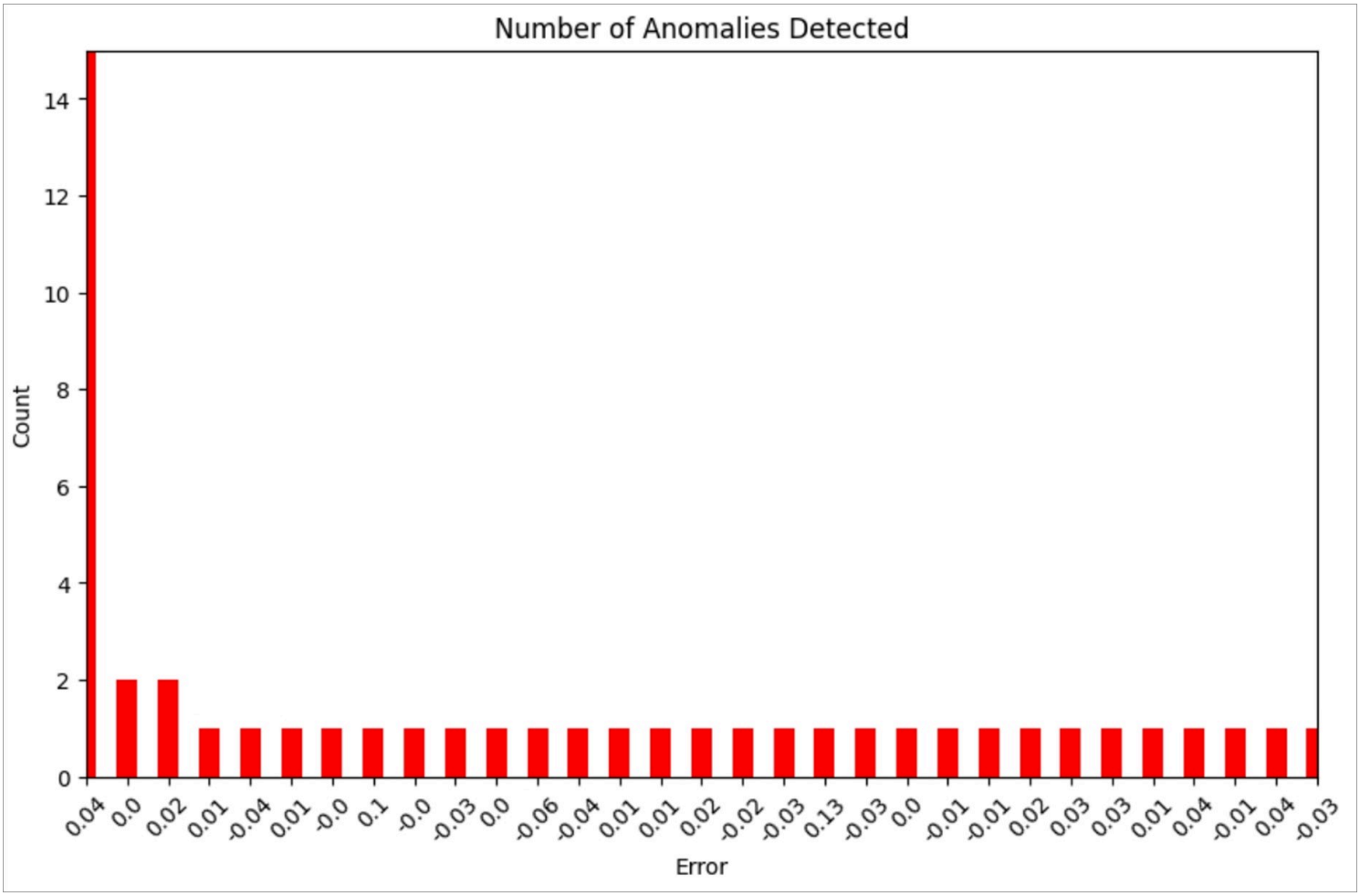


Figure 10

Number of anomalies detected on gradient boosting

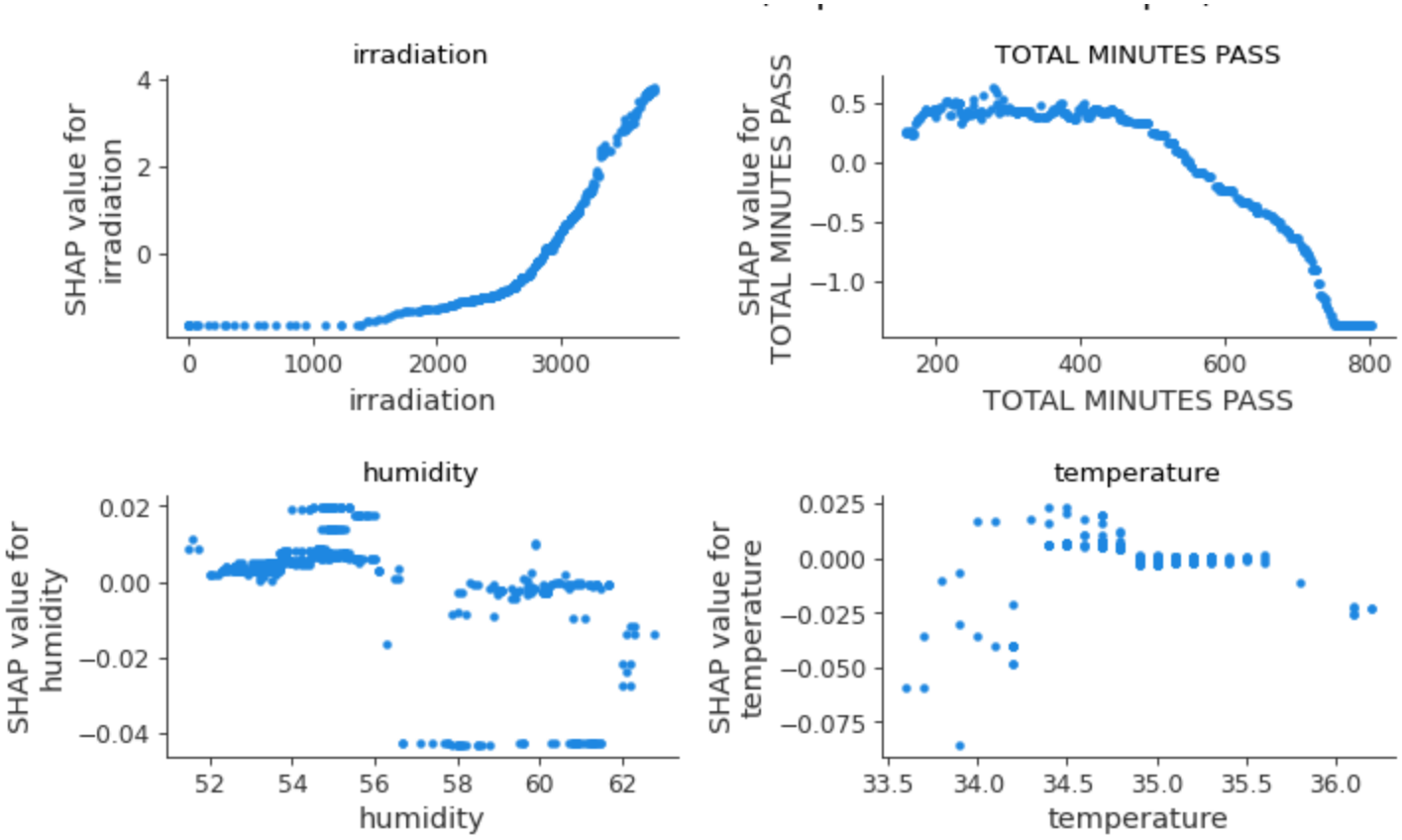


Figure 11

Dependence plot of top 4 factors affecting the prediction on gradient boosting

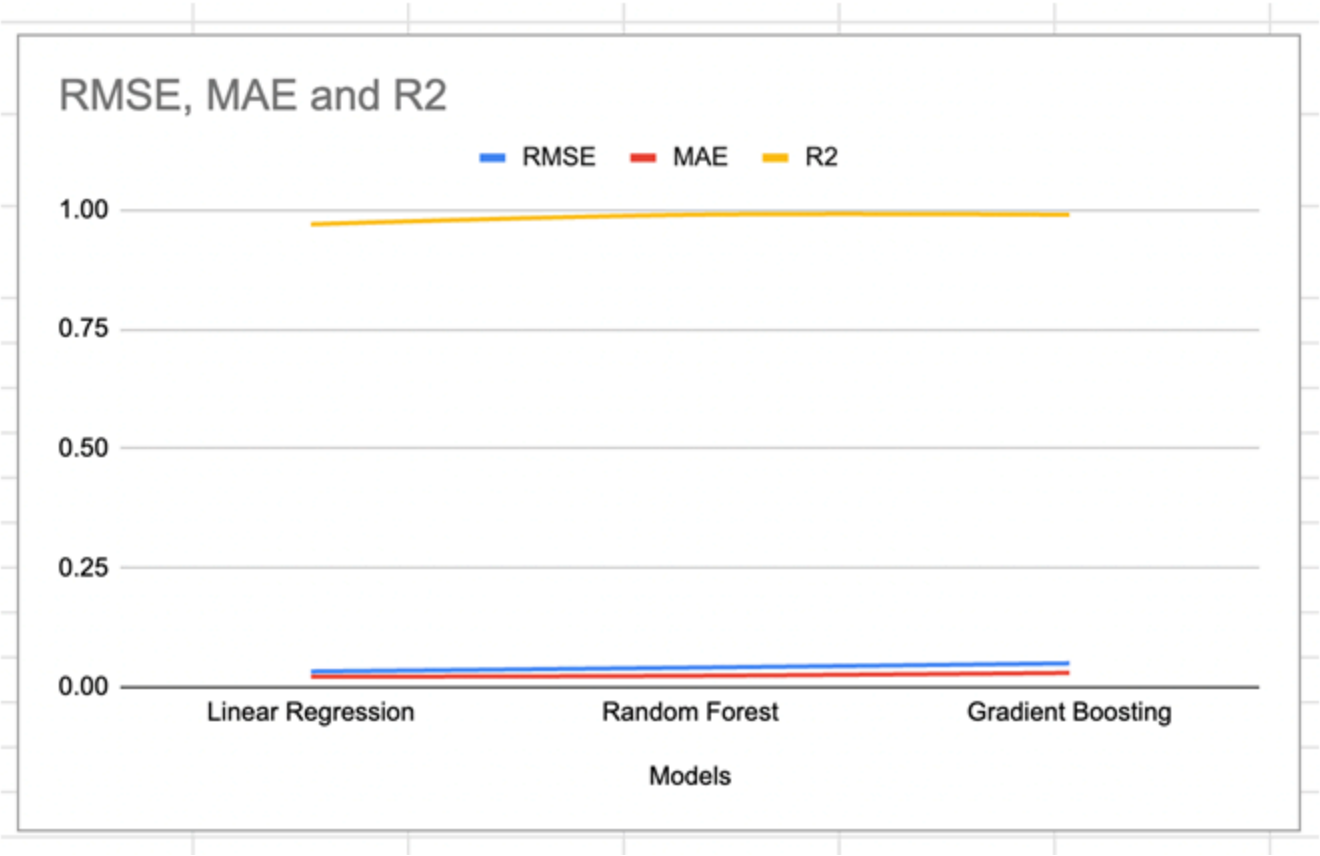


Figure 12

Comparative study of Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2) between different algorithms.

Supplementary Files

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- datasetd1.doc